**EE2001 Electrical Circuits- I:**

# Continuation of the Topic:
# Sinosoidal Steady-State Analysis

09 January 2024

# Complex Power



- Complex power is the complex sum of real power and reactive power.

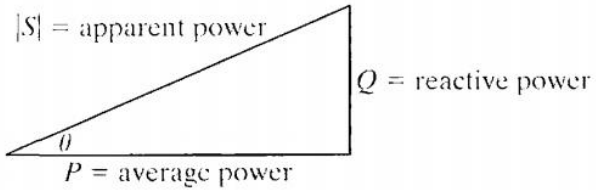
$$S = P + jQ$$

- We can compute the complex power directly from the voltage and current phasors for a circuit. The real part of the calculated value is active power and the imaginary part of the calculated value is reactive power.
- For complex power, we use the units volt-amps (VA).
- Considering that the complex power is vectoral sum of active power and reactive power all three forming a power triangle, the power factor angle $\theta_v - \theta_i$ can be calculated as below:

$$\frac{Q}{P} = \frac{(V_m I_m/2)\sin(\theta_v - \theta_i)}{(V_m I_m/2)cos(\theta_v - \theta_i)} = \tan(\theta_v - \theta_i) = tan\theta$$

- The maginute of complex power is referred to as apparent power: $|S| = \sqrt{P^2 + Q^2}$

| Quantity | Units |
|---|---|
| Complex power | volt-amps |
| Average power | watts |
| Reactive power | var |

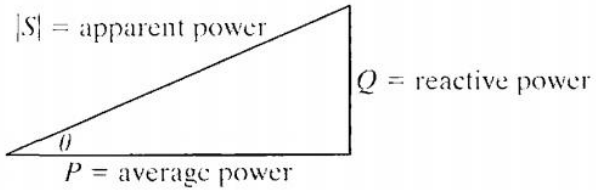


# Complex Power

4

- Although active power represents the useful output of the energy-converting device, the apparent power represents the volt-amp capacity required to supply that active power. As you can see from the power triangle given below, unless the power factor angle is 0° (that is, the device is purely resistive, pf=1, and Q=0), the VA capacity required by the device is larger than the active (real, average) power used by the device. It makes sense to operate devices at a power factor close to 1.
- Many useful appliances (refrigerators, fans, air conditioners, fluorescent lighting, washing machines) and most industrial loads operate at a lagging power factor. The power factor of these loads is corrected either by adding a capacitor to the device itself or by connecting capacitors across the line feeding the load. The latter is often used for large industrial loads.

| Quantity | Units |
|---|---|
| Complex power | volt-amps |
| Average power | watts |
| Reactive power | var |



# Complex Power



Example: An electrical load operates at 240 V rms. The load absorbs an average power of 8 kW at a lagging power factor of 0.8.

a) Calculate the complex power of the load.

b) Calculate the impedance of the load.

Solution: a) $P = |S|cos\theta$ and $Q = |S|sin\theta$

Because $cos\theta = 0.8$, $\theta = cos^{-1}(0.8) = 36.87°$ (positive because power factor is lagging) and $sin\theta = 0.6$

Therefore,

$$|S| = \frac{P}{cos\theta} = \frac{8\ kW}{0.8} = 10\ kVA$$

$$Q = |S|sin\theta = 10.\sin(36.87°) = 6\ kVAr$$

And

$$S = 8 + j6\ kVA$$

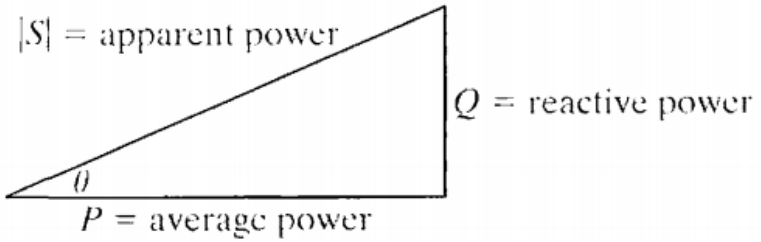
# Complex Power

4

Example: An electrical load operates at 240 V rms. The load absorbs an average power of 8 kW at a lagging power factor of 0.8.

a) Calculate the complex power of the load.

b) Calculate the impedance of the load.




Solution: b) $S = 8 + j6\ kVA$

From the computation of the complex power of the load, we see that $P = 8$ kW

$$P = V_{eff} I_{eff} cos(\theta_v - \theta_i) = (240) I_{eff} (0.8) = 8000\ W$$

$$I_{eff} = 41.67\ A$$

Calculation of the magnitude of the load impedance

$$|Z| = \frac{|V_{eff}|}{|I_{eff}|} = \frac{240}{41.67} = 5.76$$

Hence $Z = 5.76\angle 36.87°\ \Omega = 4.608 + j3.456\ \Omega$

# Power Calculations

5

- If the phasor current and voltage are known at a pair of terminals, the complex power associated with that pair of terminals is either one half of the product of the voltage and the conjugate of the current,

$$S = \frac{1}{2}\boldsymbol{V}\boldsymbol{I}^*$$

- or the product of the rms phasor voltage and the conjugate of the rms phasor current.

$$\begin{aligned} S &= V_{eff} I_{eff} \angle(\theta_v - \theta_i) \\ &= V_{eff} I_{eff} e^{j(\theta_v - \theta_i)} \\ &= V_{eff} I_{eff} e^{j\theta_v} e^{-j\theta_i} \\ &= \boldsymbol{V}_{eff} \boldsymbol{I}_{eff}{}^* \end{aligned}$$

- Note that $\boldsymbol{I}_{eff}{}^* = I_{eff} e^{-j\theta_i}$ follows from Euler's identity

$$\begin{aligned} I_{eff} e^{-j\theta_i} &= I_{eff} \cos(-\theta_i) + j I_{eff} sin(-\theta_i) \\ &= I_{eff} \cos(\theta_i) - j I_{eff} sin(\theta_i) = \boldsymbol{I}_{eff}{}^* \end{aligned}$$

# Power Calculations

5

Example: $\boldsymbol{V} = 100\angle 15°\ V$ and $\boldsymbol{I} = 4\angle -105°\ A$

$$S = \frac{1}{2}(100\angle 15°)(4\angle +105°) = 200\angle 120\,° = -100 + j173.21\ VA$$

- Once we calculate the complex power, we can read off both the real and reactive power, because $S = P + jQ$. Thus

$P = -100\ W$ and Q= $173.21\ VAr$

# Power Calculations

5

## Alternate Forms for Complex Power

- First variation is to replace voltage with the product of the current time the impedance.

$$\boldsymbol{V}_{eff} = Z\boldsymbol{I}_{eff}$$

$$S = Z\boldsymbol{I}_{eff}\boldsymbol{I}_{eff}{}^{*}$$

$$= \left|\boldsymbol{I}_{eff}\right|^{2} Z = \left|\boldsymbol{I}_{eff}\right|^{2}(R + jX)$$

$$= \left|\boldsymbol{I}_{eff}\right|^{2} R + j\left|\boldsymbol{I}_{eff}\right|^{2} X = P + jQ$$

from which

$$P = \left|\boldsymbol{I}_{eff}\right|^{2} R = \frac{1}{2} I_m{}^{2} R$$

$$Q = \left|\boldsymbol{I}_{eff}\right|^{2} X = \frac{1}{2} I_m{}^{2} X$$

# Power Calculations

5

## Alternate Forms for Complex Power

- A second useful variation is to replace the current with the voltage divided by the impedance:

$$S = \boldsymbol{V}_{eff}\left(\frac{\boldsymbol{V}_{eff}}{Z}\right)^{*} = \frac{\left|\boldsymbol{V}_{eff}\right|^{2}}{Z} = P + jQ$$

- Note that if Z is a pure resistive element,

$$P = \frac{\left|\boldsymbol{V}_{eff}\right|^{2}}{R}$$

and if Z is a pure reactive element,

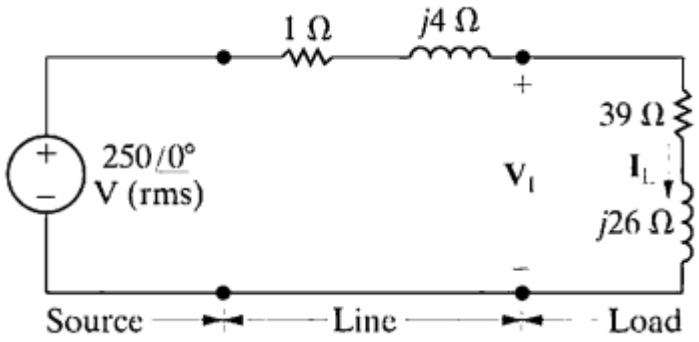
$$Q = \frac{\left|\boldsymbol{V}_{eff}\right|^{2}}{X}$$

X is positive for an inductor and negative for a capacitor.

# Power Calculations

5

Example: In the given circuit, a load having an impedance of 39+j26 Ω is fed from a voltage source through a line having an impedance of 1+j4 Ω. The effective, or rms, value of the source voltage is 250 V.




a) Calculate the load current $\boldsymbol{I_L}$ and voltage $\boldsymbol{V_L}$

b) Calculate the average and reative power delivered to the load.

c) Calculate the average and reactive power delivered to the line.

d) Calculate the average and reactive power supplied by the source.

Solution: a) The line and load impedance are in series across the voltage source, so the load current equals to voltage divided by the total impedance:

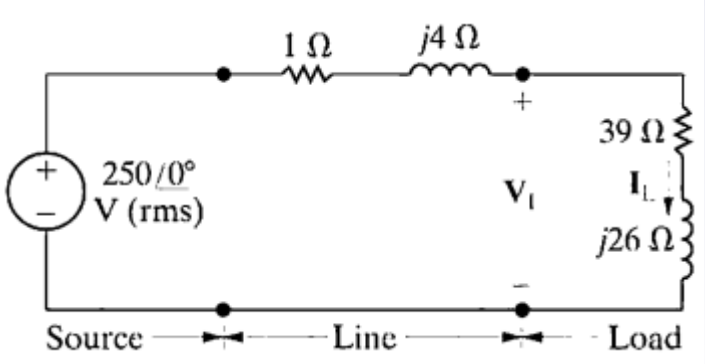
$$\boldsymbol{I_L} = \frac{250\angle 0°}{40 + j30} = 4 - j3 = 5\angle -36.87°\ A\ (rms)$$

Because the voltage is given in terms of its rms value, the current also is rms.

# Power Calculations

5

Example: In the given circuit, a load having an impedance of 39+j26 Ω is fed from a voltage source through a line having an impedance of 1+j4 Ω. The effective, or rms, value of the source voltage is 250 V.




a) Calculate the load current $\boldsymbol{I_L}$ and voltage $\boldsymbol{V_L}$

b) Calculate the average and reactive power delivered to the load.

Continuation of the solution a: The load voltage is the product of the load current and the load impedance:

$$\boldsymbol{V_L} = (39 + j26)\boldsymbol{I_L} = 234 - j13 = 234.36\angle -3.18\,^\circ\, V\ (rms)$$

b) The active and reactive power delivered to the load:

$$S = \boldsymbol{V_L}\boldsymbol{I_L}^* = (234 - j13)(4 + j3) = 975 + j650\ VA$$

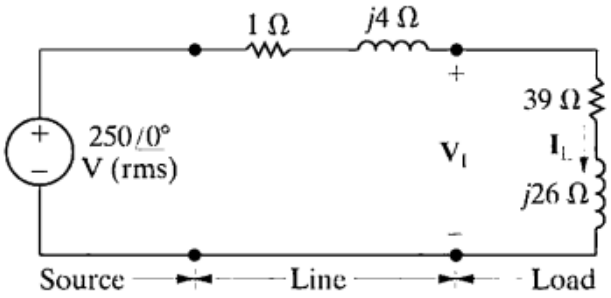
c) Becuase the line current is now known, the real and reactive power delivered to the line can be easily calculated: $P = (5)^2(1) = 25\ W$ $Q = (5)^2(4) = 100\ VAr$

Note that the reactive power associated with the line is positive because the line is inductive.

# Power Calculations

5

Example: In the given circuit, a load having an impedance of 39+j26 Ω is fed from a voltage source through a line having an impedance of 1+j4 Ω. The effective, or rms, value of the source voltage is 250 V.




c) Calculate the average and reactie power delivered to the line.

d) Calculate the average and reactive power supplied by the source.

Solution: d) One way to calculate real and reactive power delivered by the source is to add the complex power delivered to the line to that delivered to the load,

$$S = 25 + j100 + 975 + j650 = 1000 + j750\ VA$$

It can alternataively be calculated as below:

$$S_s = -250\boldsymbol{I}_L{}^*$$

The minus sign is inserted whenever the current reference is in the direction of a voltage rise:

$$S_s = -250(4 + j3) = -(1000 + j750)VA$$

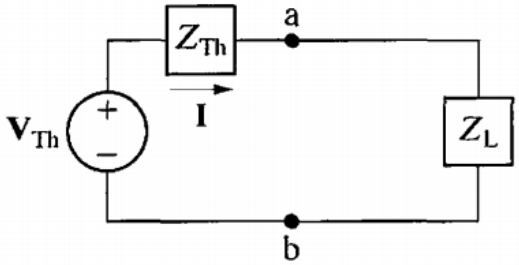
The minus sign mean that both real and reactive power are being delivered by the source.

# Maximum Power Transfer



- Remember from the previous topics that certain systems depend on being able to transfer a maximum amount of power from the source to the load. We now reexamine maximum power transfer in the context of a sinusoidal steady-state network.
- We must determine the load impedance $Z_L$ that results in the delivery of maximum active power to the load terminals.
- For maximum active power transer, $Z_L$ must equal the conjugate of the Thévenin impedance, that is

$$Z_L = Z_{Th}^*$$




expressing in rectangular form

$$Z_{Th} = R_{Th} + jX_{Th}$$
$$Z_L = R_L + jX_L$$

The active (real) power delivered to the load is

$$P = |\boldsymbol{I}|^2 R_L$$

Maximum power transfer is possible when $X_L = -X_{Th}$

# Maximum Power Transfer

6

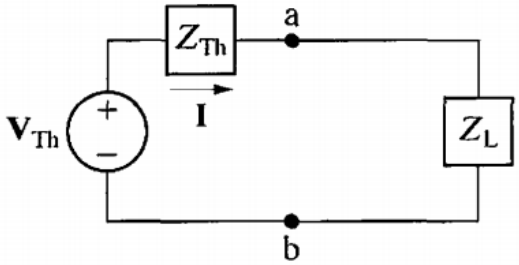
## Maximum Average Power Absorbed

The rms load current is $\boldsymbol{V}_{Th}/2\,R_L$, and the maximum average power delivered to the load is:

$$P_{max} = \frac{|V_{Th}|^2 R_L}{4{R_L}^2} = \frac{1}{4}\frac{|V_{Th}|^2}{R_L}$$

if the Thévenin voltage is expressed in terms of its maximum amplitude rather than its rms amplitude, the equation becomes

$$P_{max} = \frac{1}{8}\frac{{V_m}^2}{R_L}$$



# Maximum Power Transfer

6

## Maximum Power Transfer When Z is Restricted

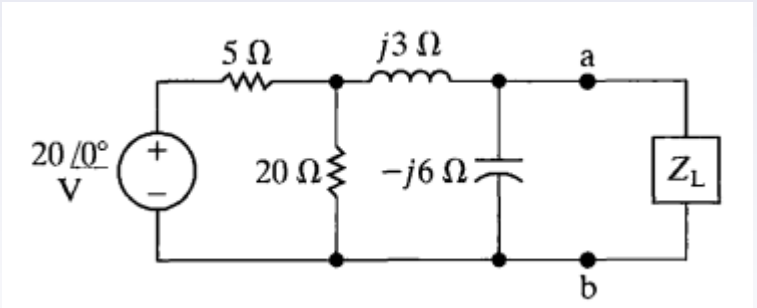
- There may be situations in which $Z_L$ can not be set equal to the conjugate of $Z_{Th}$.
- First $R_L$ and $X_L$ may be restricted to a limited range of values. In such a case, the optimum condition for $R_L$ and $X_L$ is to adjust $X_L$ as near to $-X_{Th}$ as possible and then adjust $R_L$ as close to $\sqrt{{R_{Th}}^2 + (X_L + X_{Th})^2}$ as possible.
- A second type of restriction occurs when the magnitude of $Z_L$ can be varied but its phase angle cannot. Under this restriction, the greatest amount of power is transferred to the load when the magnitude of $Z_L$ is set equal to the magnitude of $Z_{Th}$ that is when,

$$|Z_L| = |Z_{Th}|$$

# Maximum Power Transfer

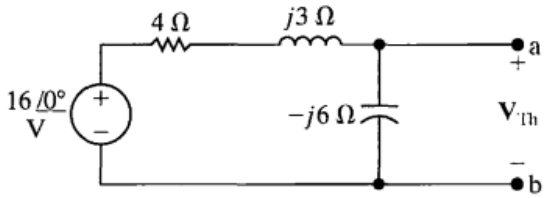
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Example: a) For the given circuit, determine the load impedance $Z_L$ that results in maximum real power transferred to $Z_L$.




b) What is the maximum average power transferred to the load impedance determined in (a)?

Solution: a) We begin by determining the Thévenin equivalent with respect to the load terminals a, b. After two source transformations, involving the 20 V source, the 5 Ω resistor and the 20 Ω resistor, we simplify the circuit:




Then, $V_{Th} = \frac{16\angle 0^\circ}{4+j3-j6}(-j6) = 19.2\angle -53.13^\circ$

$= 11.52 - j15.36\ V$

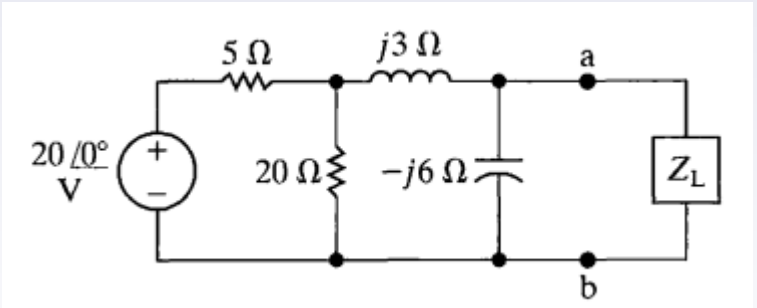
We find the Thévenin impedance by deactivating the independent source and calculating the impedance seen looking into the terminals a and b., Thus

$$Z_{Th} = \frac{(-j6)(4+3j)}{4+j3-j6} = 5.76 - j1.68\ \Omega$$

# Maximum Power Transfer

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Example: a) For the given circuit, determine the load impedance $Z_L$ that results in maximum real power transferred to $Z_L$.



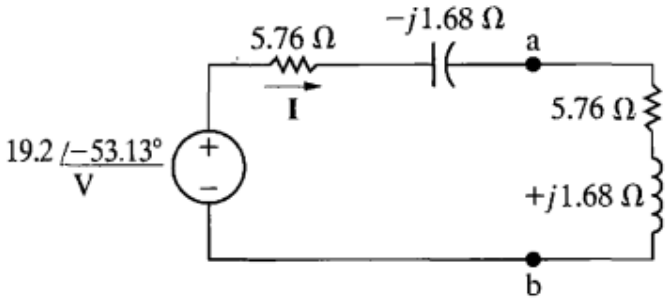

b) What is the maximum average power transferred to the load impedance determined in (a)?

Solution: a) $Z_{Th} = \frac{(-j6)(4+3j)}{4+j3-j6} = 5.76 - j1.68\ \Omega$

For maximum average power transfer, the load impedance must be conjugate of $Z_{Th}$, so

$$Z_L = 5.76 + j1.68\ \Omega$$

b) We calculate the maximum average power delivered to $Z_L$, using the circuit in which we replaced the original network with its Thévenin equivalent.
The rms value of the load current ***I***




$$I_{eff} = \frac{19.2/\sqrt{2}}{2(5.76)} = 1.1785\ A \qquad P = {I_{eff}}^2(5.76) = 8\ W$$

# End of the Lesson (09.01.24)

## Any questions?

**New supplementary videos are added to the playlist on Youtube:**

**Supplementary videos**: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist